- *Price Discovery*. As we discuss in Chapter 2 (Market Microstructure) the growth of algorithms and decline of traditional specialists and market marker roles has led to a more difficult price discovery process at the open. While algorithms are well versed at incorporating price information to determine the proper slicing strategy, they are not yet well versed at quickly determining the fair market price for a security.

CHANGING TRADING ENVIRONMENT

The US equity markets have experienced sweeping changes in market microstructure, rapid growth in program trading, and a large shift to electronic trading. In 2001, both the New York Stock Exchange (NYSE) and NASDAQ moved to a system of quoting stocks in decimals (e.g., cents per share) from a system of quoting stocks in fractions (e.g., 1/16th of a dollar or "teenies"). As a consequence, the minimum quote increment reduced from $0.0625/share to $0.01/share. While this provides investors with a much larger array of potential market prices and transactions closer to true intrinsic values, it has also been criticized for interfering with the main role of financial markets, namely, liquidity and price discovery.

The decrease in liquidity shortly after decimalization has been documented by Bacidore, Battalio, and Jennings (2001), NASDAQ Economic Research (2001), and Beesembinder (2003). This was also observed in the US equity markets after moving from a quoting system of 1/8th ($0.125/share) to 1/16th (0.0625/share) and in Canada when the Toronto Stock Exchange moved from a system of 1/8th ($0.125/share) to nickels ($0.05/share). For example, see Harris (1994, 2003), Jones and Lipson (1999), and Goldstein and Kavajecz (2000).

Some market participants argued that actual market liquidity did in fact remain stable after decimalization, although it was spread out over a larger number of price points. For example, suppose that prior to decimalization the best offer for a stock was 5000 shares at $30.00. However, after decimalization the market offers were 500 shares at $29.98, 1000 shares at $29.99, 2000 shares at $30.00, 1000 shares at $30.01, and 500 shares at $30.02. In this example, neither the total liquidity or average offered price for 5000 shares has changed, but the measured liquidity at the best market ask has decreased from 5000 shares pre-decimalization to 500 shares post-decimalization. So, while market depth (e.g., "transaction liquidity") measured as the total quantity of shares at the national best bid and offer (NBBO) has decreased, actual market liquidity may be unaffected. What has changed in this example is that now it takes five times as many transactions to fill the 5000 share order. Thus, even if liquidity has remained